

packages/graph-view/src/components/KnowledgeGraphView.tsx

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Overview

A new `KnowledgeGraphView` component is added at `packages/graph-view/src/components/KnowledgeGraphView.tsx`. It renders a force-directed graph with `@xyflow/react`, using custom node rendering, edge styling, and layout logic defined in the file.

Key changes

- **Imports** (lines 1-14): added React hooks (`useMemo` , `useCallback`) and all core `@xyflow/react` components (`ReactFlow` , `ReactFlowProvider` , `Background` , `BackgroundVariant` , `Controls` , `MiniMap`).
- **Custom node support** (lines 13-14): imports `CustomNode` and its data type `CustomNodeData` .
- **Layout utilities** (lines 17-18, 38-96): brings in `applyForceLayout` , `NODE_WIDTH` , `NODE_HEIGHT` from `../utils/layout` ; defines `getNodeDimensions` (lines 38-45) and `computeLayout` (lines 52-94).
- **Graph filtering & memoization** (lines 97-286): `KnowledgeGraphViewInner` uses `useDashboardStore` to access graph state, filters nodes by type, and memoizes layout and visual nodes/edges to avoid unnecessary recomputation.
- **Edge styling** (lines 24-35, 199-234): `EDGE_STYLES` maps edge types to CSS properties; edge rendering logic applies dynamic styles based on selection and focus.
- **Provider wrapper** (lines 288-294): default export wraps the inner component in `ReactFlowProvider` .

Impact

- **Correctness**: node sizing (`getNodeDimensions`) and stable positions (`computeLayout`) are now defined.
- **Performance**: layout and node/edge construction are memoized; only graph or filter changes trigger recomputation.
- **Maintainability**: node/edge styling and layout logic are centralized; future changes can be made in one place.
- **Compatibility**: requires `@xyflow/react` CSS import (`@xyflow/react/dist/style.css`) and the `FlowFitOnResize` helper; ensure these dependencies are present.

Risks & follow-ups

- **Layout accuracy**: verify that `applyForceLayout` produces expected positions; test with graphs of varying size.
- **Edge style fallbacks**: `EDGE_STYLES` defaults to `related` for unknown types; confirm all used edge types are covered.

- **Store integration:** confirm that `useDashboardStore` provides `graph` , `selectedNodeId` , etc., and that updates propagate correctly.
- **CSS conflicts:** the global `@xyflow/react` styles may clash with existing styles; run visual regression tests.